Data Breach Breakdown (DBB) Phases – Summary

---

1. Digital Forensic Investigation (DFI) Phases & Relation to NIST SP 800-86

Digital Forensic Investigation (DFI) follows four standardized phases defined by **NIST SP 800-86**:

DFI Phases

1. **Collection**
    - Identify, collect, and preserve potential digital evidence.
    - Ensure integrity is not compromised.
2. **Examination**
    - Extract relevant data from collected evidence.
    - Maintain forensic soundness.
3. **Analysis**
    - Interpret extracted data to answer **5WH (What, Who, When, Where, Why, How)**.
4. **Reporting**
    - Document procedures, tools, and findings.

Relation to the Proposed Framework

- The proposed framework **adopts NIST DFI structure**.
- It **enhances Examination and Analysis phases** using:
    - Data Breach Breakdown (DBB)
    - Artifact correlation
    - 5WH mapping
        - This improves **depth, structure, and completeness of investigation**

2. Data Breach Breakdown (DBB) Phases

DBB models the lifecycle of a data breach into four phases:

Phase	Investigative Focus
Infiltration	How attacker gained initial access
Propagation	How attacker moved laterally
Aggregation	How attacker accessed/collected data
Exfiltration	How attacker transferred data out

Key Insight:

- DBB is **specific to data breaches**, unlike general cyberattack models.
- Helps investigators **reconstruct attack flow logically and efficiently**

3. Selected Existing Frameworks (Table 1 Summary)

CFFTPM

- **Strength:** Time-efficient, structured phases, Windows artifact focus
- **Limitation:** Artifact classification unclear; limited scope

SRDFIM

- **Strength:** Detailed examination techniques (ASCII/non-ASCII, hidden files)
- **Limitation:** Focused on cybercrime; lacks structured 5WH mapping

IDFPM

- **Strength:** Pattern-based detection using hashes and past datasets
- **Limitation:** Cannot identify artifact or incident type solely from hash values

D4I

- **Strength:** Artifact categorization + mapping to Cyber Kill Chain (CKC); graph-based correlation
- **Limitation:**
    - CKC includes irrelevant phases (e.g., reconnaissance)
    - Windows-centric
    - Sequential (less flexible)

Overall Gap Identified

- Lack of:
    - Data breach specificity
    - Multi-source artifact integration
    - Complete 5WH mapping

4. Proposed Framework

Key Differences from D4I

Aspect	**D4I**	**Proposed Framework**
Model	CKC (7 phases)	DBB (4 phases)
Scope	Windows-focused	Multi-source (Host, Network, Security devices)
Process	Sequential	Iterative & dynamic
Analysis	Limited flexibility	Trigger-based adaptive analysis

Framework Components (Fig. 3)

1. **Evidence Categorization**
    - Sources:
        - Host (H)
        - Network Devices (ND)
        - Security Devices (SD)
2. **Content Analysis**
    - Trigger-based, iterative process:
        - Identify source
        - Extract artifacts
        - Map to DBB
        - Repeat until complete

3. **Chain of Artifacts (CoA)**
    - Links artifacts across DBB phases
    - Represented as:
        - 4D array
        - Graph
        - Linked structure
4. **Timeline Analysis**
    - Orders events chronologically → answers **"When"**
5. **Attack Flow Analysis**
    - Reconstructs attack steps → answers **"How"**
6. **5WH Mapping (Table 2)**
    - Maps findings into:
        - What, Who, When, Where, Why, How

Core Contribution

- Provides **complete, structured answers to 5WH**, which many frameworks fail to achieve

5. Case Study Summary (Tables 3–9, Fig. 5)

Scenario

- Organization: 300-employee company
- Attack type: **Spear phishing + malware**
- Trigger: IoCs from National CERT

DBB Phase Execution

Infiltration

- Malicious email sent to user (Alice)
- Attachment executed → malware infection

Propagation

- Exploitation (Zerologon vulnerability)

- Creation of unauthorized admin account ("sidious")
- Lateral movement (PC1 → OPC1)

Aggregation

- Access to file server (FS)
- Sensitive data identified

Exfiltration

- Data compressed (RAR)
- Transferred externally
- Logs partially deleted

Key Findings

- Timeline identified exact breach window
- Chain of Artifacts reconstructed full attack path
- Attack flow clearly explains sequence of compromise

- Framework successfully answered all **5WH questions**

6. Five Critical Questions

1. How does DBB improve efficiency compared to CKC in data breach investigations?
2. What challenges arise when correlating artifacts across hosts, network devices, and security devices?
3. How does iterative content analysis ensure completeness, and what are its limitations?
4. Can the Chain of Artifacts (CoA) scale effectively for large enterprise environments?
5. How will cloud and IoT environments impact the applicability of DBB-based frameworks?